IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – PathFinder — An Agentic Career Counseling Companion

Domain of Project – Education

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Recommended font style and size:

Font Style: Arial
Font Size (Heading): 28
Font Size (Content): 20

Problem Statement -

Students struggle to make informed career decisions because guidance is fragmented, self-awareness of academic strengths is limited, and industry trends shift faster than traditional counseling can track. Manual counseling doesn't scale and often misses real-time labor market signals…leading to career mismatches.

The Objective
Build an **AI-powered Career Counseling Agent** that delivers **personalized career guidance and a step-by-step roadmap** for students navigating academic and career decisions. The solution should:

- **Personalized Career Roadmaps** – Recommend courses, certifications, and skill-building steps based on academic background, interests, and goals.
- **Grounded Career Knowledge** – Draw on the Granite model's knowledge of industry trends, roles, and in-demand skills.
- **Conversational Assessment** – Ask natural follow-up questions to understand the student's strengths, interests, and constraints.
- **Proactive Guidance** – Help students course-correct early, before mismatches turn into missed opportunities.
- **Built With** –
- **LangFlow** (orchestrates the conversation flow)
- **IBM watsonx.ai** (hosts and serves the model)
- **IBM Granite model** (generates grounded, conversational responses)
- **Structured system prompt** (encodes career-counseling expertise and tone)
- **Iterative Refinement** – The agent refines its roadmap conversationally as the student shares more context.

Proposed Solution -

PathFinder is a conversational AI career counseling agent built on **LangFlow + IBM watsonx.ai + Granite models** that engages students in natural conversation to assess their academic performance, interests, and skill gaps, and generates a personalized career roadmap grounded in the Granite model's knowledge of industry trends.

How it works, in one conversation:

1. **Understands the student** — asks about academic background and self-reported skills, identifying strengths and gaps
2. **Surfaces genuine interests** — follow-up questions uncover what the student actually enjoys, not just their grades
3. **Applies career knowledge** — draws on the Granite model's built-in knowledge of in-demand skills, roles, and industry trends
4. **Builds the roadmap** — synthesizes the conversation into a step-by-step career path (courses, certifications, projects, timelines)

All four responsibilities are handled by a single, carefully prompt-engineered agent rather than separate components — a deliberate design choice for reliability within the project timeline. Multi-agent orchestration and RAG-grounded market data are planned as future

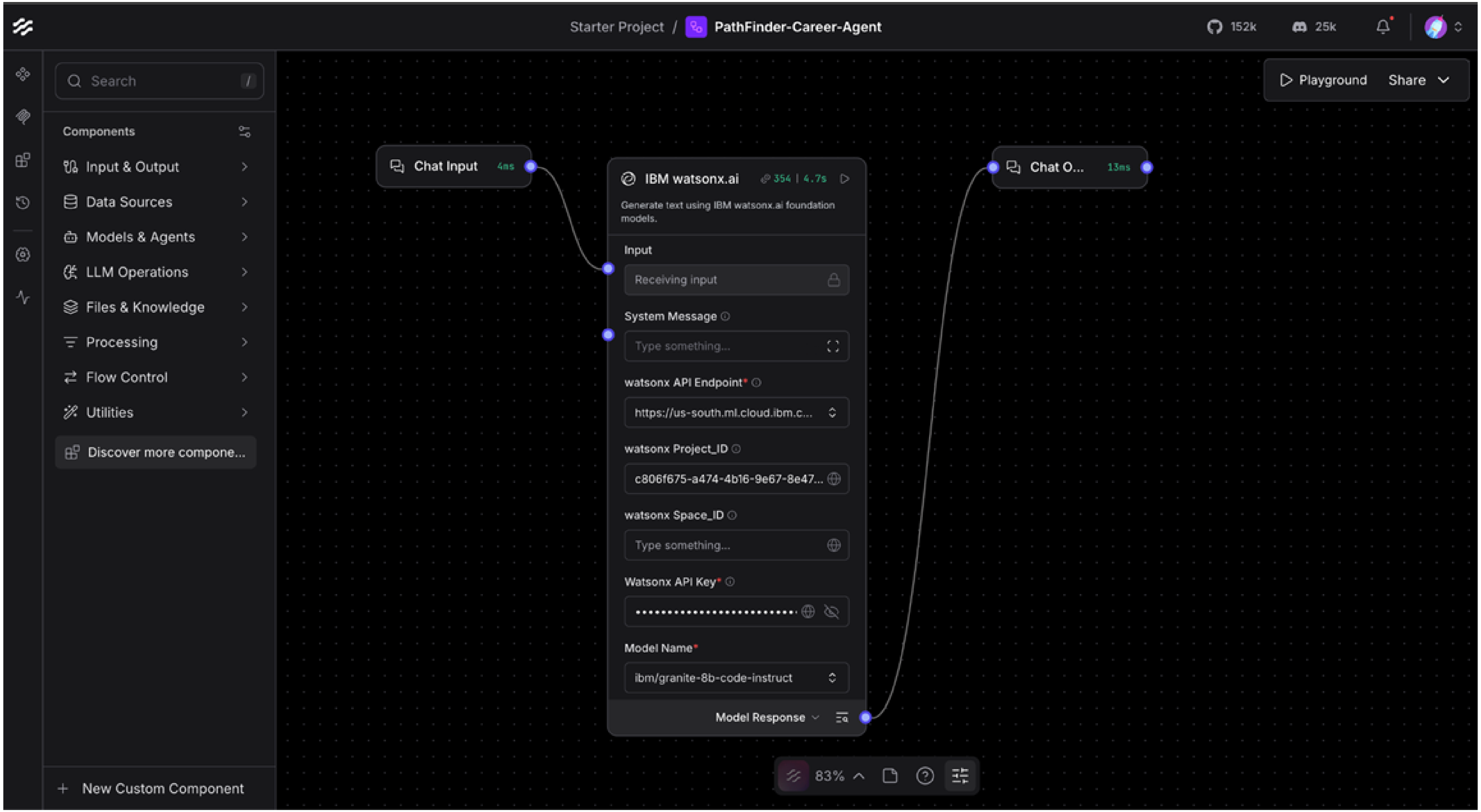
Technology Used

LangFlow (agent orchestration)
IBM Granite model — e.g. `granite-3-2-8b` or `granite-4.0-8b-instruct`
IBM Cloud + watsonx.ai (model hosting and API access)
Prompt engineering (structured system prompt encodes career-counseling logic)
LangFlow Playground (testing and validating agent responses)
GitHub (public repository for code, docs, and submission files)

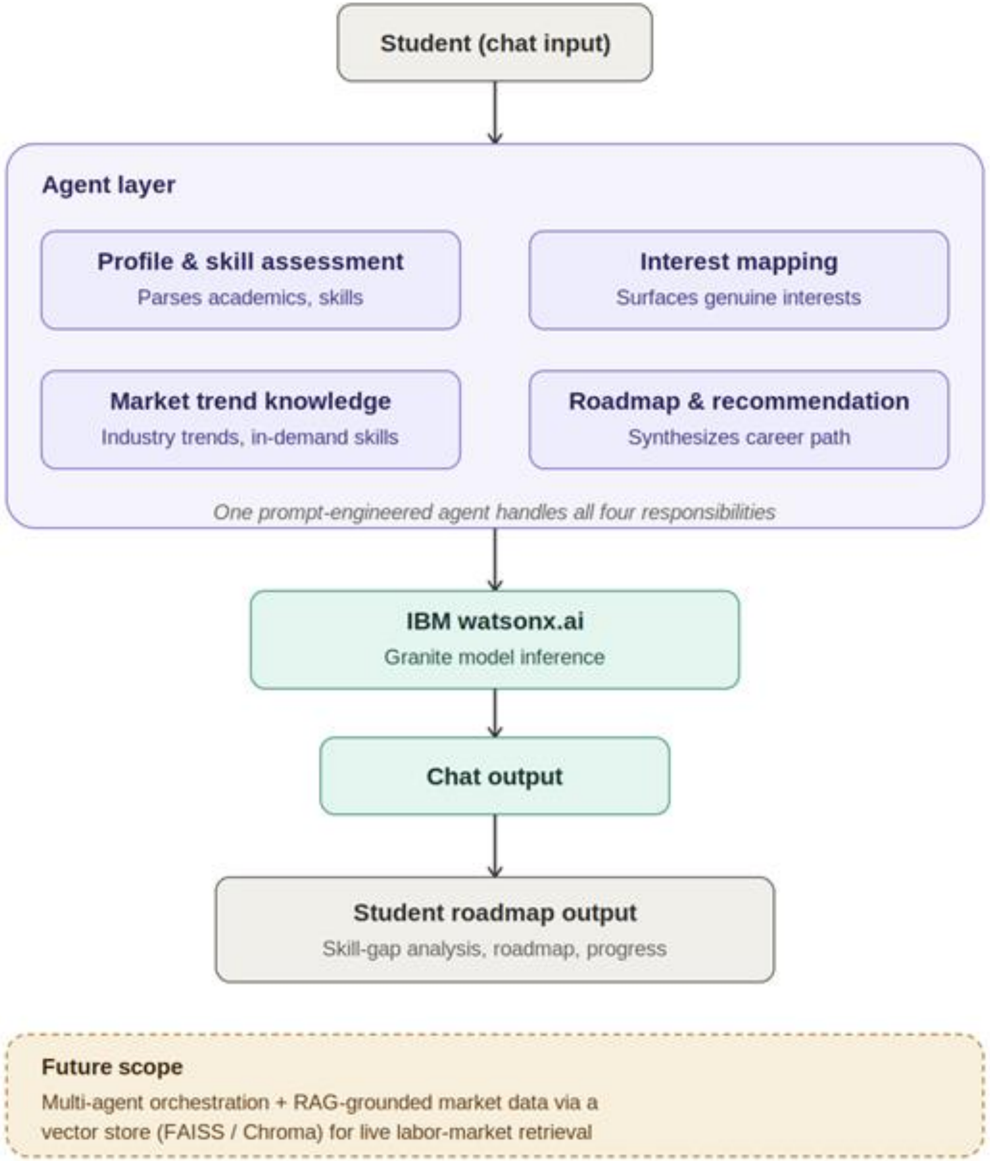
Langflow component Used -

Chat Input — student enters academic info, interests, questions
IBM watsonx AI Agent — routes to Granite model
Granite model — e.g. `Granite-4.0-8B-Instruct`
Chat Output — roadmap/recommendations displayed

Langflow workflow -



Architecture blueprint



Role of Agentic AI in the solution

Role of Agentic AI in PathFinder

Agentic AI enables PathFinder to function as an intelligent, goal-driven counselor rather than a simple chatbot. Built on **IBM watsonx.ai** and the **IBM Granite model**, it is guided by a structured system prompt that encodes four distinct responsibilities — understanding the student, surfacing genuine interests, applying career knowledge, and building a roadmap which is within a single, carefully engineered conversation.

This **prompt-driven design** ensures the agent stays focused and consistent across a multi-turn conversation, rather than answering each question in isolation.

Agentic AI also brings **context awareness**, tracking what the student has already shared the academic background, interests, constraints and so recommendations stay grounded in that specific student's situation, not generic advice.

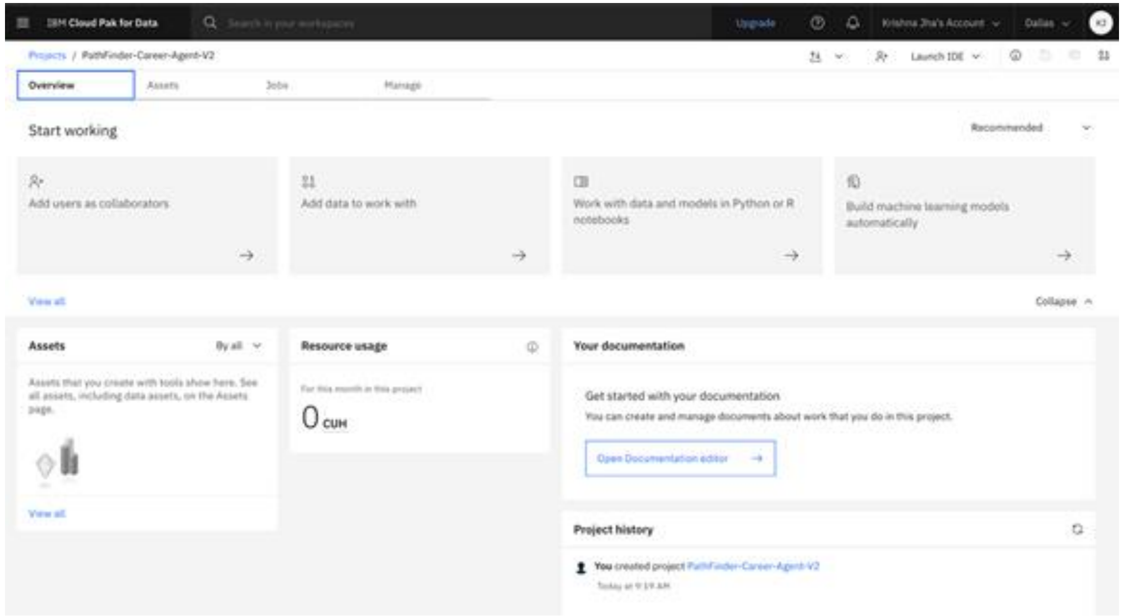
It supports **conversational adaptation**, refining the roadmap as the student provides more detail or pushes back on a suggestion, much like a real advising session.

Overall, Agentic AI makes the system **proactive, personalized, and goal-driven**, acting like a **dedicated digital career counselor** that helps students turn uncertainty into a concrete plan.

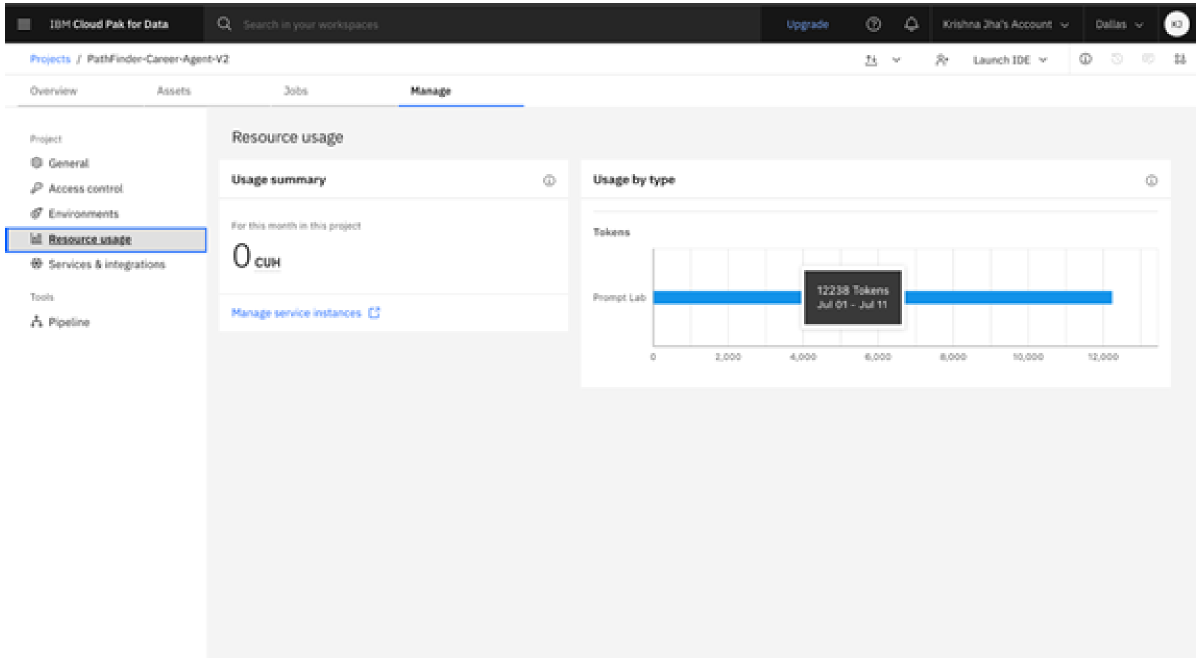
Token Usage

12,238 token are used

Before Prompt



After the Prompt



Novelty and Uniqueness

PathFinder stands out by combining **agentic prompt engineering, conversational depth, and enterprise-grade infrastructure** into a single intelligent counseling system.

Single-Agent Efficiency — Rather than juggling multiple components, one carefully engineered agent handles assessment, interest-mapping, career knowledge, and roadmap-building — simpler to build, test, and deploy reliably.

Conversational Depth — Uses natural, multi-turn dialogue to understand a student's academic background and interests, instead of a static form or questionnaire.

Grounded Career Reasoning — Draws on the Granite model's knowledge of industry trends and in-demand skills to keep recommendations realistic rather than generic.

Iterative Roadmaps — Refines its suggestions conversationally as the student shares more context, rather than issuing a one-shot recommendation.

Built for Education Access — Designed for students in fragmented or under-resourced guidance systems, where a persistent AI counselor can fill a real gap.

Scalable & Enterprise-Ready — Built on **IBM watsonx.ai** with **IBM Granite models**, ensuring reliability and governance.

This makes the solution a **focused, conversational career companion**, not just a static recommendation form — with clear headroom to grow into a multi-agent, RAG-grounded system (see Future Scope).

Git Hub Link

https://github.com/Kr1491/pathfinder-career-agent

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Future Scope

1. Multi-Agent Orchestration & RAG-Grounded Market Data:

The current single-agent design can be expanded into a multi-agent system, with dedicated agents for profile assessment, interest mapping, and market trends. A retrieval pipeline (FAISS or Chroma) would ground recommendations in live labor-market data, job postings, in-demand skills, and salary trends instead of relying solely on the Granite model's built-in knowledge.

2. Integration with LinkedIn & Job Portal APIs:

Connecting to LinkedIn, Naukri, or government labor-market APIs would let PathFinder surface real, current job listings and internship opportunities alongside its roadmap, keeping recommendations tied to what's actually hiring.

3. Regional-Language Support:

Adding support for regional Indian languages would make PathFinder accessible to students in rural and non-English-medium backgrounds which directly extending its reach to the underserved communities it's designed to help.

Certificates Earned:

Please add your credly certificate

Credly Links:
https://www.credly.com/badges/acba3ef3-3b8c-4751-81a9-25f1516dc380/public_url
https://www.credly.com/badges/c454d8d3-5fbc-4c9f-a4e9-5990eb662a33/public_url
https://www.credly.com/badges/ee176279-c613-41fc-acb0-f85b931b09f7/public_url

Certificates Earned:

Add your IBM BOB Certificate

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This certificate is presented to

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According to the Adobe Learning Manager system of record

Completion date: 23 Jun 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.